

Procurement Specification

Labeling and Bar coding Requirement

1.0 PURPOSE:

This procedure defines the labeling and bar coding requirements of incoming products.

2.0 SCOPE:

This specification applies to incoming products delivered from suppliers and is a supplement to the released component part specification and all requirements must be met in addition to the other requirements of the component specifications.

3.0 PRODUCT REQUIREMENTS:**3.1. Supplier Responsibility & Supplier Label Approval Process**

The following paper is mandatory for supplier. It is intended to attain a standard label and to ensure traceability, material flow and information flow from suppliers to customers. Prior to implementation, the supplier shall submit label samples for approval. Strict adherence to the specifications in this paper will increase benefits for our suppliers and Advantek.

It is the responsibility of supplier to provide barcode labels that meet all of the specifications listed in this document. The supplier shall be responsible for verifying the accuracy of the label and ensuring that the data is current and correct.

3.1.1 The process to follow for approval of barcode labels

1. Supplier completes request for barcode receiving label approval form (Form 11040) and submits a sample label using actual order data to Advantek.
2. Advantek will review the sample labels and provide approval or areas for correction.
3. Before supplier can begin shipping parts with the barcode label, supplier must receive written approval by Advantek.
4. Should the supplier not be able to supply according to the guideline, Advantek has to be informed in advance and the deviation to the specification approved. Failure to fully comply with this Label Specification from suppliers will be regarded as a serial supplier's capacity failure and might lead to business termination.
5. Subsequent deliveries should carry labels of equal quality to the approved labels. For every reject found at Advantek, a SCAR will be issued and supplier performance will be impacted.

3.2 Labeling and Bar Coding General Requirements

All Direct Materials shipped to Advantek requires labels with bar coding. Bar coding is required to interface with the Advantek material managing and manufacturing process. The bar code will be used as a tool for material control and for material tracking.

3.2.1 Optical and Quality Requirements

Labels supplied to this specification will meet the requirements of ISO 15416 Grade 3 alternatively ANSI X3.182 / ANSI UCC5 Grade B or better. The label must be white and printed with black in color. The adhesive glue shall not damage or harm the product by outgassing substances.

The barcode label must be permanent adhesive on the outer box. For reusable outer packing box, any old label must be removed before a new label is pasted on. Do not overlap the labels. It is recommended to use thermal transfer labels, laser printable paper or white polyester film with maximum 0.001-inch (0.025-mm) thick instead of paper labels to avoid any contamination of the direct materials by paper dust.

3.3 Label Size Requirement

Packing Size	Label Size
Normal Box	101.6mm x 101.6mm (4"x4")
Narrow Box	152.4mm x 50.8mm (6"x2")

3.4 Positioning

The placement of label on the outer box should be visible, on smooth surface and allow the barcode to be scanned. The label placement should be affixed at a standardized location, and should not be placed at / near to the packing carton / box openings.

Example 1: Label Placement on Box / Carton

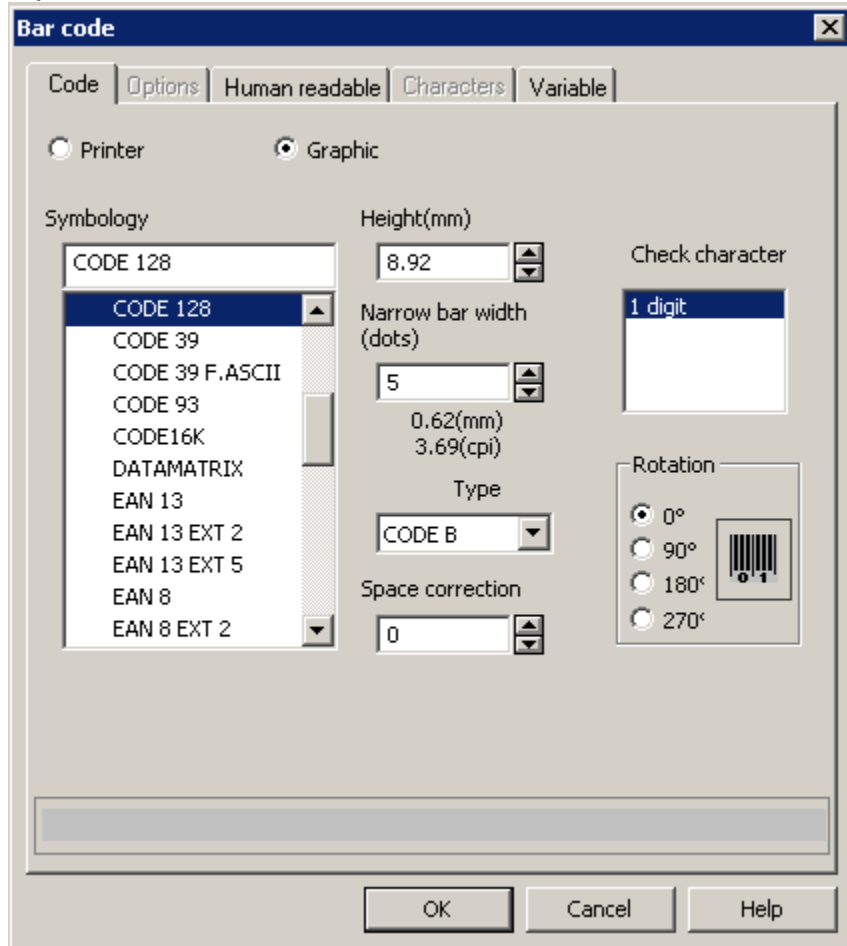


3.5 Barcode Definitions

The barcode pattern is “Code 128” with the following requirements:

The Bar height must be **min. 6 mm** or **the nearest possible measure to that**, depending on the printing method used.

Example 1:



Bar code

Code Options Human readable Characters Variable

☐ Printer ☒ Graphic

Symbology

- CODE 128
- CODE 128
- CODE 39
- CODE 39 F.ASCII
- CODE 93
- CODE16K
- DATAMATRIX
- EAN 13
- EAN 13 EXT 2
- EAN 13 EXT 5
- EAN 8
- EAN 8 EXT 2

Height(mm) 8.92

Narrow bar width (dots) 5

0.62(mm)
3.69(cpi)

Type CODE B

Space correction 0

Check character 1 digit

Rotation

- ☒ 0°
- ☐ 90°
- ☐ 180°
- ☐ 270°

OK Cancel Help

3.5 Data Contents

	Data Field	Definition	Max Length	Format	Barcode	Human Readable
1	Product Family	Specified Advantek Product Family Name	30	Alphanumeric	-	-
2	Pallet ID	Specified Advantek 12 characters	12	Alphanumeric	-	-
3	QTY	QTY is the total number of parts of the amount contained in the packing carton	4	Numbers	Code-128 Type B	In the left side of the barcode
4	UoM	UoM based on the given definition from Advantek	3	Capital Letter	-	-
5	Advantek Part Number	Specified Advantek product part number	30	Alphanumeric	Code-128 Type B	On top of the barcode
6	Item Name	Specified main product description	60	Alphanumeric	-	-
7	Lot Number	Specified supplier's manufacture lot code if supplier lot code is less than 8 characters, please fill in leading zeros.	8	Alphanumeric	Code-128 Type B	Below of the barcode
8	MFG Date	Specified item manufacturing date	10	MM/DD/YYYY	Code-128 Type B	Below of the barcode
9	EXP Date	Specified item expiry date per product shelf life	10	MM/DD/YYYY	Code-128 Type B	Below of the barcode
10	Barcode-2D	Specification refers to section 3.7	-	Alphanumeric	Data Matrix	-

3.6 Advantek Pallet ID

Advantek Pallet ID is a **12-digit unique identifier**. This number is composed by 2 segments as below:

- 2-digit Supplier Code
- 10-digit Unique sequence number from 0000000001 to 9999999999

Example 1: Advantek Pallet ID

CA0000000001

CA: 2-digit Supplier's Code

0000000001: 10-digit sequence number for the same lot. 0000000001 to 9999999999

3.7 2D Barcode

In order to collect all the printed information on the label, a Data Matrix barcode is required on the label. It needs to meet ISO/IEC 16022:2000 and ISO/IEC 24720:2006 (ISO International Standard).

3.7.1 2D Barcode requirement for Associated Products

Data fields from 2 to 11 in section 3.5 requires to be encrypted into the 2D barcode **in sequence and use vertical bar (|) as the delimiter**. If there is no value for optional data fields, please put a **space character** as the value

All the data encrypted in the 2D barcode should be in ASCII only. No double byte character.

Here is sample to show how those data fields should be encrypted in the barcode variable:

| |Pallet ID|QTY|UoM|Advantek Part Number|Item Name| |Lot Number|MFG
Date|EXP Date| | |

Example 1: Data extracted by scanning a 2D barcode with all data fields

| |CA0000167431|0020|RO|XX0092|COVER,ANTISTAT| |12345679|06/02/12|
06/02/14| | |

Note: A space character should be filled between two vertical bars for those fields without data.

Example 2: Data extracted by scanning a 2D barcode with optional data fields, , space character should be filled between two vertical bars.

| | CA0000167431 | 0020 | RO | XX0092 | | | 12345679 | 06/02/12 |
06/02/14 | | |

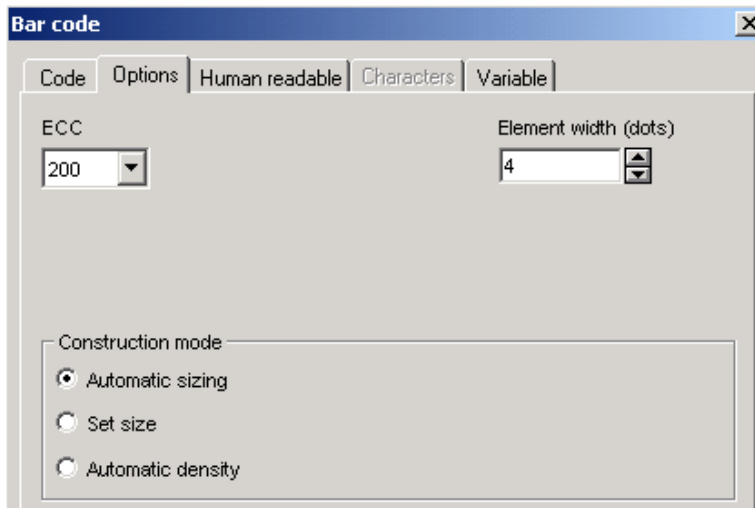
3.7.2 2D Barcode requirement for Incoming Raw materials

- 2D barcode contains |item number|batch number|

Example 1: Data extracted by scanning a 2D barcode with all data fields

|SJKEIM10RCC25|SJ090412-03|

ECC 200 is required for data correction, Element width is required at least 4 dots.



Bar code

Code Options Human readable Characters Variable

ECC 200

Element width (dots) 4

Construction mode

- ☒ Automatic sizing
- ☐ Set size
- ☐ Automatic density

3.8 Font Type and Size

Label Size : 4 X 4

Label/Text Field	Font Size	Data	Font Size	Barcode Label	Barcode Size (height X width)	Type
Product Name	5.03					
Pallet ID	4.03	Pallet ID	5.03	Pallet ID	7.05 X 5	Code B
QTY	4.03	QTY	5	QTY	6.04 X 4	Code B
Advantek Part Number	4.03	Advantek Part Number	5.03	Advantek Part Number	6.04 X 5	Code B
Description	4.03					
Lot	4.03			Lot	6.04 X 3	Code B
Mfg Date	4.03			Mfg Date	6.04 X 4	Code B
Exp Date	4.03			Exp Date	6.04 X 5	Code B
UoM	4.03	UoM	5.03			

Refer to the given sample below.

For 4" x 4" label:



Label Size : 6 X 2

Label/text Field	Font Size	Data	Font Size	Barcode Label	Barcode Size (height X width)	Type
Product Name	5.03					
Pallet ID	4.03	Pallet ID	5.03	Pallet ID	7.05 X 5	Code B
QTY	3			QTY	5.02 X 3	Code B
Advantek Part Number	4.03	Advantek Part Number	5.03	Advantek Part Number	6.04 X 5	Code B
Description	4.03					
Lot	3			Lot	5.02 X 3	Code B
Mfg Date	3			Mfg Date	5.02 X 3	Code B
Exp Date	3			Exp Date	5.02 X 3	Code B
UoM	3	UoM	5.03			

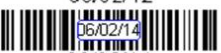
Refer to the given sample below.

For 6" x 2" label:


3.9 Barcode Label Example

3.9.1 Normal Outer Box Label: Label Size 101.6mm x 101.6mm (4"x4")

Example: (Not in scale)

AX Cover Tape	
Pallet ID: CA0000167431	
	
QTY: 0020	UoM: RO
Advantek Part Number	
XX0092-300	
	
Description	
COVER, ANTISTAT, 9.2MM, 300M	
LOT: 	
MFG Date: 	
EXP Date: 	

3.9.2 Normal Outer Box Label: Label Size 152.4mm x 50.8mm (6"x2")

Example: (Not in scale)

AX Cover Tape		QTY: LOT: MFG: EXP:				
Pallet ID: CA0000167431						
		0020	12345679	06/02/12	06/02/14	
Advantek Part Number		UoM:				
XX0092-300		rolls				
						
Description						
9.2mm, 300mm						
Cover Tape, Anti-Stat 9.2mm, 300mm						

Document History Log

Revision	Date	CCD#	Description of Change	Originator
3.0	9/4/12	None	1. Change Barcode Code-128 Type C to Barcode Code – 128 Type B 2. Remove “Advantek Inventory Number field 3. Add Pallet ID – Section 3.6 4. Specified Bar code requirement for AP and Raw material 5. Add Table of Requirement of Labels both for 4”x4” and 6”x2” 6. Change Label samples from Proband to Covertape 7. Increase Font size of Human Readable (Lot, Mfg date and Exp date)	J. de Mesa
2.0	4/26/11	None	1. Remove <i>“Both left and right margins (Silence Zone) must be 8.0 mm wide minimum from the beginning to end”</i> from Section 3.5 2. Remove Barcode sample from Section 3.5 8. Modify Type code from B to C on Section 3.5 Contents 9. Add “24-digit unique identifier on Section 3.6 10. Add a new section, Section 3.7 – Supplier Lot number 11. Revise wordings on Section 3.8	J. de Mesa
1.0	03/29/11	None	New	J. de Mesa

SUPPLIER ACKNOWLEDGEMENT AND CONFORMANCE CERTIFICATE

This is to certify that _____, have read and understood all requirements that
(Name of Company)
Advantek, Inc. had stated in the Procurement Specification. Having viewed Procurement Specification
for the Labeling and Barcoding Requirement.

_____ will be responsible for all requirements and ensure conformance to this
(Name of Company)
Procurement Specs and hereby agrees to the requirements as of _____:
(Date)

1. Supplier has received a copy of Procurement Specification for Labeling and Barcoding Requirement.
2. Supplier will take all the necessary measures to ensure that the Procurement Specifications is disseminated to all appropriate departments and personnel within Supplier's organization, such that these requirements of the Procurement Specifications are adhered to all relevant levels;
3. Supplier acknowledges that this Procurement Specification may be modified and updated from time to time, and agrees to take all appropriate measures to ensure that supplier remains in compliance with the most current revision of the Procurement Specification at all times;
4. Supplier will immediately notify the appropriate Purchasing representative and Quality representative at Advantek of any failure on the part of the Supplier to comply with the latest revision of the Procurement Specification;
5. This Certificate will remain effective until revoked in writing by Supplier.

Supplier Company Name

Company Director / General Manager